

PHISHING EMAIL DETECTION & AWARENESS REPORT

Cyber Security Internship — Task 2

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Organization:

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Project Title:

Phishing Email Detection & Awareness System

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OBJECTIVE

The objective of this investigation is to analyze suspicious email samples and identify phishing-related indicators used in phishing attacks.

This report aims to classify email risks, understand how phishing attacks manipulate users through social engineering techniques, and create cybersecurity awareness to help users recognize and prevent phishing attempts.

SCOPE

This investigation includes the analysis of phishing and legitimate email samples using header analysis, URL and domain investigation techniques.

The investigation focuses on identifying phishing indicators, analyzing suspicious email behavior, classifying email risk levels, and providing cybersecurity awareness recommendations including prevention tips and employee do's and don'ts.

TOOLS USED

- Public Phishing Email Samples

(used for phishing email analysis and investigation)

- Google Message Header Analyzer

<https://toolbox.googleapps.com/apps/messageheader/>

(used for email header and routing analysis)

- MXToolbox Email Header Analyzer

<https://mxtoolbox.com/EmailHeaders.aspx>

(used for email header investigation and blacklist analysis)

- Browser Investigation Tools

(used for checking suspicious links, domains, and URLs safely)

- Google Docs / MS Word / PDF

(used for documentation and report preparation)

INVESTIGATION STEPS

The investigation was performed using the following steps:

1. Collection and selection of phishing and legitimate email samples
2. URL and domain investigation of suspicious links
3. Email header analysis using header investigation tools
4. Identification and analysis of Indicators of Compromise (IOCs)
5. Behavioral analysis of phishing and social engineering techniques
6. Threat intelligence investigation of suspicious domains and URLs
7. Correlation-based investigation using multiple findings
8. Final risk classification of analyzed email samples

EMAIL SAMPLE ANALYSIS

Three email samples were selected and analyzed to identify phishing behavior, suspicious indicators, and legitimate communication patterns.

Sample 1 — Safe Banking Email

Subject: Your Monthly Bank Statement

Dear Harshini,

Your monthly account statement for May 2026 is now available.

Please log in securely through the official banking portal to review your statement.

<https://www.hdfcbank.com>

Regards,

HDFC Bank Support

Classification: SAFE

Observations:

- Legitimate domain identified
- HTTPS secure connection used

- Personalized greeting observed
- No urgency or fear tactics detected

Sample 2 — Phishing Email

Subject: 🚨 Urgent: Your Account Will Be Suspended

Dear User,

We detected suspicious activity in your account.

Verify immediately to avoid permanent suspension.

<http://hdfc-secure-login-verification.xyz>

Failure to verify within 12 hours will lock your account permanently.

Security Team

Classification: PHISHING

Observations:

- The email used urgency-based language to create panic and pressure users into immediate action.
- The URL used HTTP instead of HTTPS, increasing suspicion.
- The domain structure did not match the legitimate HDFC Bank domain.
- A short verification deadline was used as a social engineering tactic.
- Generic greeting (“Dear User”) was used instead of personalized communication.

Sample 3 — Phishing Email

Subject: Netflix Payment Failed

Dear Customer,

Your recent payment failed.

Update your card immediately to continue services.

<http://netflix-billing-update.ru>

Thank you,

Netflix Billing Center

Classification: PHISHING

Observations:

- The email used urgency-based language to pressure users into immediate payment updates.
- The URL used HTTP instead of HTTPS, increasing suspicion.
- The domain structure did not match the legitimate Netflix domain.
- The suspicious “.ru” TLD increased phishing suspicion during investigation.
- The email attempted to create payment-related panic using fake billing issues.
- Generic greeting (“Dear Customer”) was used instead of personalized communication.

HEADER ANALYSIS

The following sample email header was analyzed to investigate sender information, routing details, and possible spoofing behavior.

Delivered-To: user@gmail.com

Received: from secure-login-alert.xyz (185.234.219.12)

by mx.google.com with ESMTPS id x7si1234567qkf.123.2026.05.17

for <user@gmail.com>;

Subject: Urgent Account Verification

From: Security Team <support@secure-login-alert.xyz>

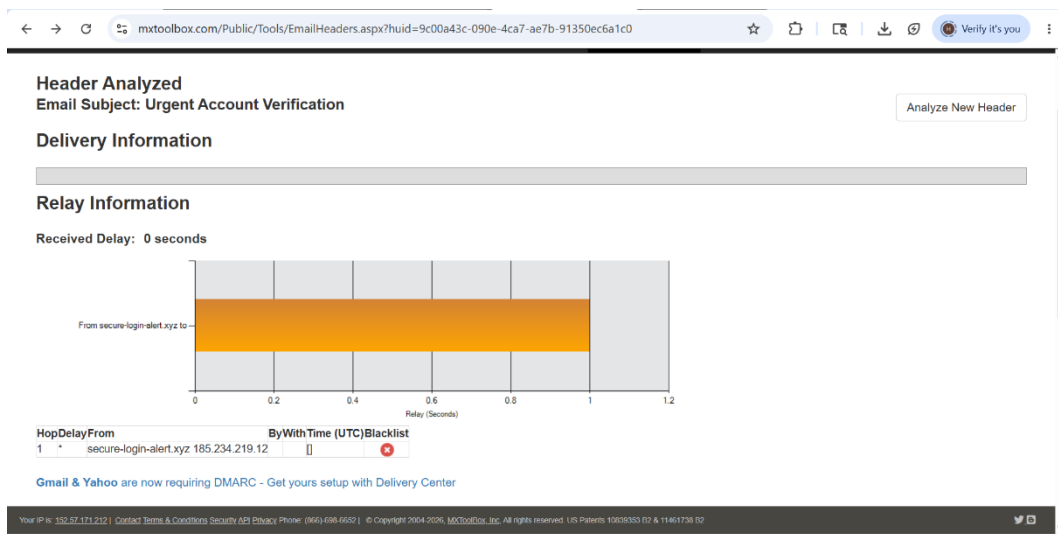
To: user@gmail.com

Date: Sun, 17 May 2026 10:15:22 +0000

Message-ID: 12345@secure-login-alert.xyz

Header analysis was performed using MXToolbox and Google Message Header Analyzer to investigate sender details, routing information, and possible spoofing behavior.

Mxtoolbox:



Findings:

- The sender domain `secure-login-alert.xyz` appeared suspicious during investigation.
- The email originated from the IP address `185.234.219.12`, which was identified as an unknown external mail server.
- MXToolbox showed blacklist warning indicators during analysis, increasing phishing suspicion.
- The domain used security-themed wording such as “secure” and “login” to imitate legitimate security communication.
- The analyzed email showed characteristics commonly associated with phishing infrastructure.

Google message header:

The screenshot shows the Google Admin Toolbox Messageheader tool interface. The header bar is green with the text "Google Admin Toolbox Messageheader" and a "Help" link. The main content area displays the following message details:

Messageid	12345@secure-login-alert.xyz
Created at:	5/17/2026, 3:45:22 PM GMT+5:30 (Delivered after)
From:	Security Team <support@secure-login-alert.xyz>
To:	user@gmail.com
Subject:	Urgent Account Verification

Below the message details, there is a table showing the delivery path:

#	Delay	From *	To *	Protocol	Time received
0		secure-login-alert.xyz	→ [Google] mx.google.com	ESMTPS	5/17/2026, 3:45:22 PM GMT+5:30

Findings:

- The sender email address used the suspicious domain `secure-login-alert.xyz`.
- The email was delivered from an external sender source to Gmail infrastructure.
- Routing information was analyzed successfully using Google Message Header Analyzer.
- The subject line “Urgent Account Verification” used urgency-related wording commonly observed in phishing emails.
- The overall header structure showed possible phishing-related behavior during investigation.

URL INVESTIGATION

URL investigation was performed to identify suspicious domains, phishing-related infrastructure, and spoofing behavior used in phishing attacks.

Investigated URL 1

<http://hdfc-secure-login-verification.xyz>

Findings:

- The URL used HTTP instead of HTTPS.
- The domain did not match the legitimate HDFC Bank domain.
- The `.xyz` TLD increased phishing suspicion.
- Security-themed words such as “secure”, “login”, and “verification” were used to create false trust.
- The URL demonstrated phishing-related characteristics commonly associated with banking credential theft attacks.

Investigated URL2

<http://netflix-billing-update.ru>

Findings:

- The URL used HTTP instead of HTTPS, increasing suspicion during investigation.
- The domain structure did not match the legitimate Netflix platform domain.
- The `.ru` TLD increased phishing suspicion during analysis.
- Payment-related wording such as “billing” and “update” was used to pressure users into immediate action.

- The URL demonstrated phishing-related characteristics commonly associated with credential and payment information theft attacks.

PHISHING INDICATORS IDENTIFIED

The following phishing indicators were identified during the investigation:

- Urgency-based language used to pressure users into immediate action
- Generic greetings such as “Dear User” and “Dear Customer”
- Suspicious domains not matching legitimate organization domains
- Use of HTTP instead of secure HTTPS communication
- Suspicious TLDs including `.xyz` and `.ru`
- Brand impersonation targeting banking and streaming platforms
- Payment-related and security-themed social engineering wording
- Short verification deadlines designed to create panic
- Possible spoofing behavior identified during header analysis

IOC ANALYSIS

The following Indicators of Compromise (IOCs) were identified during the phishing investigation:

IOC TYPE	VALUE	REASON FOR SUSPICION
Domain	secure-login-alert.xyz	Suspicious security-themed spoofed domain identified during header analysis
Domain	hdfc-secure-login-verification.xyz	Fake banking-related phishing domain
Domain	netflix-billing-update.ru	Suspicious payment-related phishing domain
URL	http://hdfc-secure-login-verification.xyz	Fake banking verification URL used in phishing activity

URL	http://netflix-billing-update.ru	Suspicious payment update phishing URL
IP Address	185.234.219.12	Unknown external mail server source identified during header investigation
Sender Email	support@secure-login-alert.xyz	Suspicious sender identity associated with phishing activity

RISK CLASSIFICATION

Email Sample	classification	Reason
Sample 1 — Monthly Bank Statement	SAFE / LOW RISK	Legitimate domain, HTTPS connection, and no phishing indicators identified
Sample 2 — Account Suspension Email	PHISHING / HIGH RISK	Spoofed banking domain, urgency tactics, suspicious URL, and phishing indicators identified
Sample 3 — Netflix Payment Failed	PHISHING / HIGH RISK	Fake payment-related domain, suspicious TLD, urgency-based language, and phishing-related behavior identified

HOW THE ATTACK WORKS

The phishing emails used social engineering techniques to manipulate users into clicking suspicious links and revealing sensitive information.

The attackers attempted to create panic and urgency by using messages related to account suspension and payment failure. Users were pressured into taking immediate action without properly verifying the legitimacy of the email.

The phishing emails contained spoofed domains designed to imitate legitimate organizations such as HDFC Bank and Netflix. Suspicious URLs redirected users to fake verification or payment-related pages intended to steal credentials and financial information.

The attackers also used generic greetings, fake security warnings, and short response deadlines to increase the likelihood of user interaction.

PREVENTION STEPS

Users can reduce phishing risks by following these cybersecurity best practices:

- Verify sender email addresses carefully before responding.
- Avoid clicking suspicious or shortened links.
- Check whether websites use secure HTTPS connections.
- Never share passwords, OTPs, or banking details through email.
- Be cautious of urgent messages requesting immediate action.
- Verify suspicious requests through official company websites or support channels.
- Avoid downloading attachments from unknown senders.
- Enable Multi-Factor Authentication (MFA) for additional account security.
- Report suspicious emails to the security or IT team immediately.

EMPLOYEE DO'S AND DONT'S

Do's:

- Verify sender email addresses carefully.
- Report suspicious emails immediately.
- Use official websites for account verification.
- Enable Multi-Factor Authentication (MFA).
- Check URLs before clicking links.

Dont's:

- Do not click suspicious links or attachments.
- Do not share passwords or OTPs through email.
- Do not trust urgent messages without verification.
- Do not download files from unknown senders.
- Do not respond to suspicious security or payment requests.

CONCLUSION

This investigation successfully analyzed phishing email samples and identified multiple phishing-related indicators including suspicious domains, urgency-based language, spoofing behavior, and social engineering techniques.

The investigation demonstrated how attackers use fake verification requests and payment-related messages to manipulate users into revealing sensitive information.

The report also highlighted the importance of email awareness, safe browsing practices, and proper verification techniques to reduce phishing risks and improve cybersecurity awareness.